

@attorneighhh

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(PROBABLY ALMOST)

EVERYTHING ON THE SYLLABUS

XBI11 IAL EDEXCEL BIOLOGY

UNIT 1 & 2

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Biodiversity & Conservation

▼ What are the optimal conditions for bacterial growth?

Bacteria reproduce by binary fission where each bacterial cell splits into two about every 20 minutes.

Ideal Condition for Bacterial Growth:

- Warmth - to maximise the rate of enzyme-controlled reactions.
- Moisture - bacteria need water to dissolve nutrients in and for many chemical reactions.
- Nutrients - bacteria need various ions for survival (such as phosphate ions for ATP) and glucose for respiration, all of which they obtain from their external environment.
- Optimum pH - to maximise the rate of enzyme-controlled reactions.
- Oxygen - Required for species of bacteria that respire aerobically.

▼ How can plant substances can have anti-microbial properties? How can these be tested?

To investigate bacteria for scientific purposes it needs to be cultured. Bacteria is typically cultured on **agar plates** which have the ideal conditions needed for Bacterial Growth.

Concerns of Culturing:

- Microorganisms used may themselves be harmless but there is always the risk of mutant, pathogenic bacterial strains arising.
- Culture may be contaminated from contaminants in the env.
- Growing a pure strain of a microbe is especially difficult as microbes in the air or on the skin may contaminate it.

Method:

Precautions: Aseptic Techniques must be used:

- All equipment must be sterilised before the culture is started.
- Once culture is grown, it cannot leave the lab.
- The instrument used to inoculate the agar plate/add bacteria to the plate must be sterilised using a bunsen Burner Flame
- A yellow bunsen burner flame may be left on a lab bench to create convection currents to carry airborne microorganisms away.
- Disinfect the work area
- After use, all cultures must be disposed of safely by sealing them in plastic bags and sterilising them at high heat.

Steps:

1. Sterilise the inoculating loop in the bunsen burner flame until it glows red. Then leave to cool.
2. Dip the sterilised loop in the bacteria suspension and then streak the loop across the agar, avoiding digging into it.
3. Replace the petri dish lid, tape closed and label. Turn dish upside down.
4. Place small pieces of plant material/small filter paper soaked in plant extracts or filter paper soaked in plant based drugs on the surface of the agar with the now cultured bacteria.
5. Any clear areas of jelly around these samples indicate that bacteria are not growing and that the samples has an antimicrobial chemical in it.

▼ Drug Tests

William Withering's digitalis soup

Through **trial and error**, the scientist William Withering developed a remedy for the condition dropsy - swelling of tissue due to the accumulation of excess water, now called oedema - by using **foxglove** and an active ingredient within it known as **digitalis**. By chance he discovered that one of his patient's dropsy was treated when using an older remedy that contained foxglove and from there made different versions of the remedy containing different concentrations of foxglove, this was his '**digitalis soup**'.

Too much digitalis resulted in him **poisoning** his patients, but too little had **no effect** on the dropsy, it was through this trial and error that Withering found the **optimum dosage**, something that is found today by starting with very small doses that slowly increase and monitoring of the effects it has.

Stages of Drug Testing

1. The drugs are tested on **human tissues and cells** grown in a lab and **computer models** predict the effect of them.
2. The drugs are then tested on **animals**. UK law states each drug must be tested on at least **2 different species**; it is done to view the side effects of the drugs.
3. The final main stage is **clinical trials** which involve testing the drugs on human volunteers and is made of 3 main phases:
 - **Phase 1** - Testing on a small group of **healthy volunteers (100-300)** - this again checks for **side effects** and is monitored to see the areas of the body it effects and how long it remains in the body for.
 - **Finding are then reviewed by independent {scientists / medics} to see if work can progress to stage 2**
 - **Phase 2** - Testing the drug on a **small group of volunteers with the illness (100 - 300) groups of patients**. This is done to start determining **effectiveness** in both treating and preventing the disease and also to **find the optimum concentration for dosage**.
 - **Phase 3** - Once a drug has passed phase 2 it is tested on **a large number of participants (groups of 1000-3000)**, with the disease. This final stage is done using a double blind test, placing participants randomly in 2 groups: with one group receiving the treatment and the other one receiving a placebo. This is done to confirm **effectiveness, dosage, side effects, longer-term effects** of and to analyse results to see if there is a significant difference with and without treatment.

A placebo group -

- **Placebo** - Substance with **no active ingredients**; it does not contain the drug being tested and is normally just a sugar pill.
- When patients are taking medication there is what is known as '**the placebo effect**' where the patient's condition improves because **psychologically** they believe the drug will make them better, so they do make some improvement.

- Drug testing need to compare to the placebo group of patients to observe whether the drug is **more effective than the psychological effects**.

Double-blind trials -

- A **blind** trial is where the patient does not know whether they have the active drug or the placebo drug; this is so the placebo effect can be observed.
- A **double-blind** trial is where **neither the patient nor the doctor** knows which drug they have; this is to stop any **subconscious bias** the doctor may have when assessing the patient and their improvement.

Peer review -

- This is where studies such as drug testing are **reviewed** by other professionals to be sure they're **accurate, well carried out and significant**. This **avoids public scares** such as the one created over the MMR vaccine and autism, which are based on poorly conducted studies that aren't representative of the drug or medicine in question.

Classification

▼ Key Terms:

- **Classification:** The means of organising the variety of life based on relationships between organisms using differences and similarities in phenotypes and in genotypes.
- **Biodiversity:** Measure of the variety of living organisms and their genetic differences in a given area
- **Taxonomy:** The science of classifying, describing and naming living organisms.
- **Species:** Groups of organisms sharing similar characteristics which can interbreed/reproduce to produce fertile offsprings.
- **Molecular Phylogeny:** The use of genetic material analysis and comparing nucleotide base sequences of genes and amino acid sequences to establish evolutionary relationships between organisms
- **Endemism:** Species that are found only in one place on earth and nowhere else
- **Species Richness:** The number of different species in an area

▼ Outline the Means of Classification

1. Artificial Classification based on Human Judgement: Organising groups of organisms based on visible physical characteristics (morphology)
2. Molecular Phylogeny/Phylogenetic Classification: Observing homologous structures (structures showing common ancestry) and comparing nucleotide base sequences of genes and amino acid sequences to identify closely related ancestors of species.

▼ Outline the Main Taxonomic Groups

Domain, Kingdom, Phylum, Class, Order, Family, Genus, Species

(Remember King Phillip Came over for Good Soup)**

3 Domains:

- Archaea: Archaeobacteria (primitive bacteria often found in extreme environments)
- Bacteria: Eubacteria (the true bacteria we know, pathogenic bacteria, bacteria recycling nutrients in the soil, etc.)
- Eukarya: Protista, Fungi, Plantae & Animalia

From there, each domain is classified into **smaller, non-overlapping groups called taxons**. The order of taxons in the phylogenetic classification system is **domain, kingdom, phylum, class, order, family, genus and species**. All organisms are universally named using the **binomial naming system** which consists of the genus and the species, for instance humans are universally named the '*Homo sapiens*'

Biodiversity

▼ What is biodiversity and why is it important?

▼ How do you measure biodiversity?

- **Species richness** - The **number** of different species in a given area.
- **Species Abundance** - The evenness in the **distribution** of species in the area
- **Index of diversity** - The **number and type** of different species in a given area.

Formula:

Simpson's Index Continued

$$D = \frac{N(N - 1)}{\sum n(n - 1)}$$

n = the total number of organisms of a particular species
 N = the total number of organisms of all species

This version of Simpson's diversity index ranges between 1 (low diversity) and infinity, where larger numbers mean higher diversity.

If an area had 12 cows, 10 hippos and 2 monkeys, it's diversity index would be:

$$\begin{aligned} & 24(24 - 1) / 12(12-1) + 10(10-1) + 2(2-1) \\ & = 2.46 \end{aligned}$$

- Genetic Diversity within a species - Heterozygosity Index

$$\text{heterozygosity index} = \frac{\text{number of heterozygotes}}{\text{number of individuals in the population}}$$

▼ Why is biodiversity becoming threatened?

- **Deforestation** - this causes the **loss of habitat and food sources** for a number of species.
- **Over-exploitation** - this is where humans are using organisms and land **unsustainably**. Fishing for instance, is causing the deaths of some groups of fish faster than their population can be increased again by births.
- **Climate change** - the burning of fossil fuels has lead to increased emissions of greenhouse gases causing **global warming** and **climate change**. This has caused mass **habitat destruction** in the arctic causing populations of arctic animals to plummet. It has also caused the climate to change faster than animals can adapt, so **decreased their populations and changed their distribution** across the world.

- **Pollution** - pollution of chemicals, plastics and other materials across land, lakes and the seas has caused the deaths of many individuals, leading to **endangerment and extinction** of many species.

▼ Outline the Hardy-Weinberg Principle and Equation

The Hardy-Weinberg equilibrium is a principle stating that the **genetic variation in a population will remain constant from one generation to the next** in the **absence of disturbing factors** such as mutations, natural selection, non-random mating, migration, etc., when considering large populations.

Large populations as the chances of losing an allele in the small gene pool is less likely

Allele Frequency:

$p + q = 1$ (Where p = dominant allele and q is recessive allele)

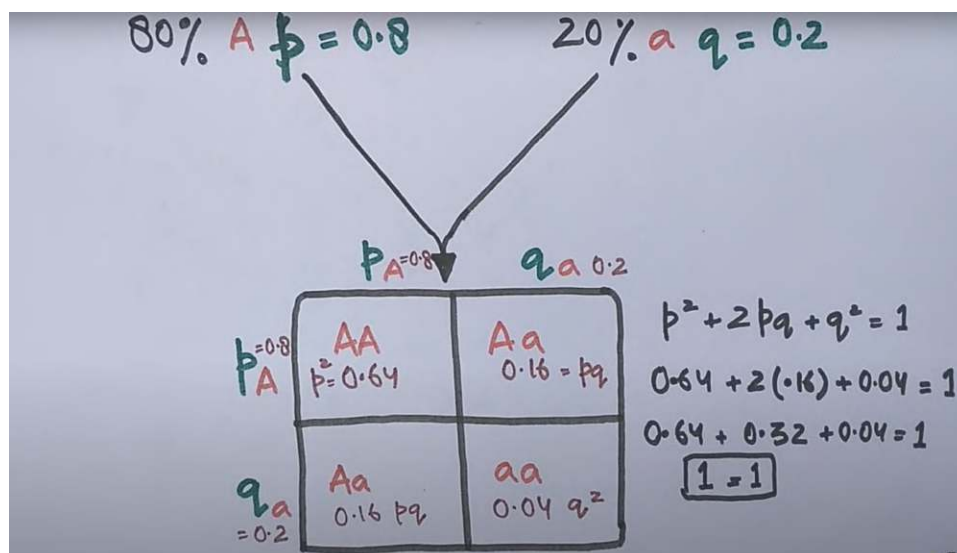
Genotype Frequency:

$(p + q)^2 = 1$ **OR** $p^2 + 2pq + q^2 = 1$

p^2 = frequency of **AA** (homozygous dominant)

$2pq$ = frequency of **Aa** (heterozygous)

q^2 = frequency of **aa** (homozygous recessive)



▼ What are the factors that change allele frequencies?

- **Mutation** - mutations in genes can result in a **new allele**, resulting in a change in allele proportions.
- **Natural selection** - Natural selection is the process in which fitter individuals, who are **better adapted** to the environment, **survive and pass on the advantageous genes to future generations**. Evolution is the process by which the **frequency of alleles in a gene pool changes over time as a result of natural selection**.

▼ Outline the process of natural selection

Evolution via natural selection:

1. There's a **variety** of phenotypes and genotypes within a population.
2. **An environmental change** occurs and as a result of that the **selection pressure changes**. Selection pressures are factors that affect an organism's chances of survival.
3. Some individuals possess **advantageous alleles** which give them a **selective advantage** and allow them to **survive and reproduce**.
4. The advantageous alleles are passed on to their **offspring**.
5. Over time, the frequency of advantageous alleles over disadvantageous alleles changes in a population and leads to evolution

▼ Outline how Reproductive Isolation leads to speciation

If two populations become reproductively isolated, they can no longer interbreed and there is no gene flow between them. New genetic information accumulates independently in each population over time, due to different environments, selection pressures, and mutations. Eventually, they become different species.

3 main changes can result in reproductive isolation:

1. Seasonal - when individuals of the same species have **different mating seasons**.
2. **Mechanical** - changes in genitalia that **physically prevent** some members of a population breeding with others
3. **Behavioural** - a group of individuals within a population could develop **courtship rituals** that aren't attractive to the main population

▼ Outline how geographical isolation leads to speciation

- Geographical Isolation happens when a physical barrier divides a population of a species i.e., floods, volcanoes, etc.,
- Conditions (Abiotic & biotic factors) are different on either side of the barrier
- This causes members on each side of the barrier to develop different characteristics that suit their different environments via natural selection in response to selection pressures
- This can be through mutations that lead to the introduction of a new allele that gives that population a selective advantage to survive. When they survive, they pass on the advantageous allele to their offsprings until the allele frequency increases.
- Over time, due to these mutations, the population on either side of the barrier will become genetically different and can no longer interbreed to produce fertile offsprings (reproductive isolation).

▼ Differentiate between Genetic Biodiversity and Species Richness

Genetic Diversity considers the variety of alleles and genotypes within a species whereas species richness considers the number of different species within a given habitat/area

Conservation of Ecosystems

▼ Outline Ecosystems

- Ecosystem: All the organisms living in a particular area (biotic factors), known as the community, as well as all the non-living elements of that particular environment (abiotic factors).
- Ecology: The complex system of interactions between the abiotic and biotic factors in an ecosystem.
- All species in an ecosystem has a specific role in its habitat known as a niche, which consists of its biotic and abiotic interactions with the environment.

▼ Why do organisms need to adapt in an ecosystem?

- Organisms adapt to their own niche so they can survive without competition. If 2 species occupy the same niche, one will be better

adapted and eventually outcompete the other species. Adaptations can be:

- **Anatomical - structural features** of an organism that increase its chances of survival. I.E. Waxy skin to cut water loss
- **Behavioural - the way an organism acts** to increase its chances of survival. For instance, animals like possums that play dead to avoid predators.
- **Psychological - processes inside an organism's body** that increase its chances of survival. For instance, bears hibernate through winter and lower their metabolic rate to conserve energy so they don't need to hunt or gather food.

▼ Evaluate the Methods used in conserving genetic diversity of endangered species

- **Scientific research** such as studying the behaviour of animals, working on improving breeding success to increase the population size and controlling and eradicating diseases that have the potential to be lethal to animals
- **Captive breeding programmes + Reintroduction programmes:**
Endangered species are **carefully bred** to increase genetic diversity and population size then released into their **natural habitat** to restore lost habitats.

Q. How can breeding programmes maintain genetic diversity? (koalas)

1. (zoos) select koalas from {north and south / different areas in} Australia ;

1. inter-zoo exchange (of animals / semen) / eq ;

2. idea of use of {stud books / record keeping} (to select mates) ;

3. reference to { preventing / eq} inbreeding ;

4. idea of avoiding genetic drift ;

5. use of {IVF/AI/surrogates};

6. process for measuring genetic diversity described ;

Cons:

- Not enough space and resources for most zoos to carry these out
- Often difficult to provide the right conditions for breeding

- Reintroduction is likely to be unsuccessful if the original cause of extinction isn't removed
- Animals bred in captivity may find it difficult to adjust to unsupported life in the wild.
- Expensive and time consuming
- **Education programmes** which aim to educate people about **the importance of maintaining biodiversity, captive breeding programmes, as well as illegal trade of animal products.**

Question Number	Answer
8 (d)	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material indicated as relevant. Additional content included in the response must be scientific and relevant. <u>On the island</u> • introducing laws to protect the albatross (D) • removal of predators (D) • through trapping / poisoning of mice / introducing a predator of mice (D) • so more chicks will survive to breeding age (E) • fishing exclusion zone (D) • supplying more food to albatross (E) • more food available for parents to give to chicks (E) • therefore increase population size / conserve the albatross (E) • reintroduction program (D) • strategies to ensure increased survival chances of reintroduced birds e.g. removal of mice, behavioural conditioning (E) <u>In zoos</u> • Tristan albatross breeding pairs taken to zoos (D) • captive breeding programmes (D) • reference to studbooks (D) • collecting eggs and taking them elsewhere to hatch (D) • therefore offspring are not eaten by mice / protected from predators (E) • offspring reintroduced to the island (D) • to increase population size (E) • reintroduce Tristan albatross currently held in zoos (D) • therefore increase population size (E) • (captive breeding/studbooks/breeding zoo albatrosses used) to {maintain / increase} genetic diversity (E) • Hardy Weinberg equation used to see change in allele frequency over time (E)

(6)

▼ Outline the use of seed banks in conservation

Seed Banks store a large number of seeds in order to conserve genetic diversity and **prevent plant species from going extinct**. Storing seeds, instead of plants, means that a large variety of species can be conserved, it's also **cheaper** than storing whole plants as it **takes up less space**. The seeds are first X-Rayed to see if they contain a developed embryo, then stored in **cool, dry conditions (between -20 and -40 degree C)** as this maximises the amount of time they can be stored for and they are periodically tested for **viability**



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